

Report

SBA 930.2

Exploring LLM Use Cases in Customer Support

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Overview

Steps

Task 1: Exploring LLM Use Cases in Customer Support

Prompt:

What are some practical applications of LLMs in enhancing customer support? Focus on how LLMs can assist with FAQs, automate responses to common inquiries, support agents with real-time information retrieval, and improve user experience through chatbots or personalized recommendations.

Machine:

LLMs can improve customer support by handling routine requests quickly, helping human agents find accurate information, and making interactions more personalized. They work best as assistants for customers and agents—not as a replacement for human judgment on sensitive or complex cases.

Task 2: Identifying Limitations

Prompt:

What limitations do LLMs face when handling customer support inquiries? Please provide specific examples.

Machine:

LLMs face several limitations in customer support. They can misunderstand context when a customer gives incomplete information; for example, if someone says, “My order still hasn’t arrived,” the model may not know which order they mean without account details. They can also provide incorrect or outdated information, such as giving the wrong return deadline if company policies have changed. LLMs may give generic or irrelevant answers to unusual issues—for example, responding with tracking advice when a customer says a package was marked delivered but was stolen. They can recognize emotional language but cannot genuinely understand empathy or make judgment calls in sensitive situations, such as a customer dealing with a medical

emergency or a fraud claim. Complex cases, legal complaints, and technical problems requiring specialized company knowledge should be escalated to a human agent.

Task 3: Crafting Effective Prompts

Prompt:

How can I formulate prompts to improve the quality of responses I get from an LLM in a customer support scenario?

Machine:

To improve LLM responses in customer support, write prompts that are specific, clear, and provide enough context. State the LLM's role, the customer's situation, the information it can use, and the exact type of answer you want. For example, instead of asking, "Help this customer," use: "You are a customer-support agent for an online electronics store. A customer says their headphones stopped charging after two weeks. Write a polite response, ask two clarifying questions, and provide basic troubleshooting steps. Do not promise a refund; offer escalation if the steps fail."

An ineffective prompt is vague, such as "Answer this customer question," because the LLM does not know the company, the issue, or the desired tone. An effective prompt sets boundaries, such as requiring a concise and empathetic tone, using only approved policy information, and asking the LLM to say when it is uncertain or needs to escalate to a human.

Task 4: Ethical Considerations in AI

Prompt: What ethical considerations should be taken into account when implementing LLMs in customer support roles?

Machine:

When implementing LLMs in customer support, companies must protect customer privacy and securely handle personal information such as names, addresses, order histories, and payment details. They should collect only necessary data, limit who can access it, and explain how customer conversations are stored or used.

Companies also need to watch for bias. An LLM could produce unfair, offensive, or less helpful responses based on a customer's language, name, location, or other characteristics. Responses should be tested and monitored so all customers receive respectful and consistent treatment.

Transparency is important: customers should be told when they are interacting with an AI system rather than a human. They should also have an easy way to reach a human agent, especially for complaints, financial issues, account security, or other sensitive situations. Human oversight is necessary to review errors, correct harmful outputs, and make final decisions in high-risk cases.

Task 5: Case Study Reflection

Prompt:

Can you summarize a successful case study of LLM implementation in customer support and what lessons can be learned from it?

Machine:

A successful example is Klarna's AI customer-service assistant, launched in 2024. Klarna reported that the assistant handled about two-thirds of its customer-service chats in its first month, completing work comparable to hundreds of full-time agents. The company also reported that average resolution time dropped from about 11 minutes to under 2 minutes, while repeat inquiries decreased because customers received faster, more complete answers.

The main lesson is that LLM support tools work best when they handle high-volume, routine questions such as payments, refunds, order status, and account issues. Companies should monitor answer quality, connect the system to accurate and current company information, protect customer data, and give customers a clear option to reach a human agent for complex or sensitive problems. The goal should be to support human agents and improve response time—not let AI make important decisions without oversight.

A challenge was maintaining accurate and helpful responses at scale, so the company needed to monitor the assistant's performance and connect it to reliable, current customer-support information.

Task 6: Summarizing Insights and Lessons

Prompt:

As we conclude, what are the key insights and lessons learned about the applications and limitations of LLMs in customer support?

Machine:

LLMs can improve customer support by answering frequently asked questions, automating routine responses, helping route tickets, and giving human agents quick access to relevant information. They can make service faster and available at any time, especially for simple requests such as checking an order status, resetting a password, or explaining a return policy.

However, LLMs can misunderstand context, give inaccurate or outdated answers, and struggle with emotional, unusual, or high-risk situations. They should be connected to reliable company information, monitored for quality, and designed to escalate complex issues—such as fraud, legal complaints, or sensitive billing disputes—to a human agent.

Conclusion

This lab showed me that LLMs can make customer support faster by handling common questions and helping agents find information. However, they should not replace people in complex, emotional, or high-risk situations. If I helped implement an LLM support system, I would make sure it uses current company information, protects customer privacy, and gives customers an easy way to reach a human agent.

Appendix